

## E-waste management in India

I read the article 'E-waste management in India' published in January issue and found it really interesting and informative.

We are planning to develop a sustainable e-waste recycling method suitable for Indian conditions and are happy to bring to your notice that we could achieve it to a good extent. Keeping maximum material recovery in mind, we have been storing several parts like small motors and some kinds of batteries which do not have proper recycling solutions at present.

We request you to publish more articles on e-waste management and related issues so as to create awareness among various electronic component manufacturers.

F.R. George  
Manager-Projects & Systems  
E-Parisaraa Pvt Ltd, Bengaluru

## PCB Design Software

I have been a regular reader of EFY since 1984 and learnt a lot from it. I have tried PCB and schematic designing software included in the EFY-CD and obtained good results as far as electronic circuits are concerned. However, I was unable to draw electrical circuits because the required components were not available in the library. Please include in the EFY-CD schematic designing software suitable for both electrical as well as electronic circuits.

Suyog Today  
Through e-mail

## Basics of Electronics

I am an electronics engineer working in government sector for about a decade but out of touch with practical electronics. Now I have started reading EFY to improve my practical knowledge. I want to know more about the components used in the electronic circuits. So please tell me the names of the relevant books.

Syed Anwar Owais  
Srinagar

EFY: We suggest you to read particularly 'Circuit Ideas' and 'Construction' sections of EFY magazine, regularly. You may also go through 'ABC of Amateur Radio and Citizen Band' by Rajesh Verma, which was published by EFY a few years ago. It also describes some very basic concepts of electronics. Incidentally, we have also started giving hands-on training on Basics of Electronics and some other subjects. Let's hope we shall soon be able to impart this training in your city as well. At present, we are doing so in NCR and Bengaluru only.

## Readers' Requests

I have been reading EFY for over nine years and request you to publish articles on:

1. LPG leakage sensors and techniques
2. Central locking system for cars
3. Wireless DC power supply
4. Electronic consoles for cars and bikes

Rajeev Kumar  
Rohtak

☐ Please publish the circuit of a microcontroller-based UPS/inverter.

Swapan Sarkar  
West Bengal

## Capacitance Meter

I have assembled the 'Microcontroller-based Capacitance Meter' project published in April 2008 issue, on a breadboard. I have used the same components as specified in the article, except for the 4-digit, 7-segment display unit, which I have replaced with four separate 7-segment displays.

There seems to be some consistency in the values displayed, but the problem is that all the six displays do not glow simultaneously. Each time I press the reset switch, only one out of the six displays activates. Also, the displays don't refresh automatically. For example, when I connected a 1 $\mu$ F capacitor in CUT (capacitor under test), the value displayed was 0783 nF. But to know this value, I had to press the reset switch a number of times. Please help.

Shreelekha  
Through e-mail

The author, Prof. K. Padmanabhan, replies:

Please check whether the connections from pins 2 and 3 are shorting with pin 1 of IC CD4066. Also check the power supply for ripples. The display may not refresh properly if decoder IC 74138 has some shorts on its pins 1, 2, 3 with pin 16. Check the AT89C2051 chip by replacing it with a freshly programmed chip.

## Heart Rate Meter

In the 'Microcontroller-based Heart Rate Meter' project published in May 2008 issue, a light-dependent resistor (LDR) is used as the sensor. LDRs are slow devices. Nowadays, sensitive photo-transistors and Darlington pair photo-transistors are readily available. These are fast and sensitive compared to LDRs. I think the sensitivity of the project can be increased by using these devices.

Pradeep G.  
Siliconics, Bengaluru

The author, Prof. K. Padmanabhan, replies:

Certainly, you can use any photo-transistor or photo Darlington such as 2N5777. Here, the transient performance is not a criterion since the change in light due to blood flow through the capillaries (which is used for finding the heart beat) is so slow that even an LDR is quite fast for it.

Further, the LDRs are available in various sizes, ranging from 4 mm to 2 cm in diameter. We've used an LDR with 1cm dia here. An LDR having a larger surface area is recommended to detect all the scattered light passing through your thumb finger. If you use a photo-transistor, only those few rays which can fall upon the tiny spot of the base of the transistor (<1 mm in size) will be the active light (relating to the change in blood flow.) This will be lesser than the amount of light that an LDR (1cm) can collect from the pulp of the thumb finger. If a small lens is used in front of the photo-transistor, it will perform better.